

Gradient Descent Optimizes Over-parameterized Two-Layer ReLU Networks: A Neural Tangent Kernel Analysis

1 Preliminaries

Notation. We write $[n] = \{1, \dots, n\}$. For a vector v , $\|v\|$ is its Euclidean norm; for a matrix M , $\|M\|_2$ is the operator (spectral) norm and $\|M\|_F$ the Frobenius norm. For a symmetric matrix M , $\lambda_{\min}(M)$ denotes its smallest eigenvalue. Throughout, $c, C, C_1, C_2, \dots$ denote universal positive constants whose values may change from line to line and depend only on quantities specified in each statement's hypothesis; constants that depend on a problem parameter carry it as a subscript (e.g. C_{λ_0}). The training data are $\{(x_i, y_i)\}_{i=1}^n$ with inputs normalized to $\|x_i\| = 1$ and labels $|y_i| \leq 1$.

Model and training. We study the two-layer ReLU network

$$f_W(x) = \frac{1}{\sqrt{m}} \sum_{r=1}^m a_r \sigma(w_r^\top x), \quad \sigma(z) = \max\{z, 0\}, \quad (1)$$

with first-layer weights $W = (w_1, \dots, w_m) \in \mathbb{R}^{d \times m}$ and output weights $a_r \in \{\pm 1\}$. At initialization $w_r(0) \sim \mathcal{N}(0, I_d)$ i.i.d. and $a_r \sim \text{Unif}\{\pm 1\}$ i.i.d., independent of the $w_r(0)$. Only the first layer is trained: the a_r are fixed at their initial values and gradient descent updates W on the squared loss

$$L(W) = \frac{1}{2} \|y - u(W)\|^2, \quad u_i(W) := f_W(x_i), \quad (2)$$

with the update rule, for step size $\eta > 0$,

$$W(k+1) = W(k) - \eta \nabla_W L(W(k)), \quad k = 0, 1, 2, \dots \quad (3)$$

We abbreviate $u(k) := u(W(k))$ and write the residual as $y - u(k) \in \mathbb{R}^n$.

Definition 1.1 (Empirical and population NTK Gram matrices). For weights W , the *empirical Gram matrix* $H(W) \in \mathbb{R}^{n \times n}$ has entries

$$H_{ij}(W) = \frac{1}{m} \sum_{r=1}^m x_i^\top x_j \mathbf{1}\{w_r^\top x_i \geq 0\} \mathbf{1}\{w_r^\top x_j \geq 0\}. \quad (4)$$

The *population NTK Gram matrix* $H^\infty \in \mathbb{R}^{n \times n}$ has entries

$$H_{ij}^\infty = \mathbb{E}_{w \sim \mathcal{N}(0, I_d)} \left[x_i^\top x_j \mathbf{1}\{w^\top x_i \geq 0\} \mathbf{1}\{w^\top x_j \geq 0\} \right] = \mathbb{E}[H_{ij}(W(0))]. \quad (5)$$

We abbreviate $H(k) := H(W(k))$. Both matrices are symmetric positive semidefinite.

Assumption 1.2 (Spectral gap of the NTK). The population Gram matrix is strictly positive definite:

$$\lambda_0 := \lambda_{\min}(H^\infty) > 0.$$

Remark 1.3. Assumption 1.2 is the *single* spectral assumption of the analysis; every constant below is tracked as a polynomial in $1/\lambda_0$. It is mild: $\lambda_0 > 0$ holds whenever no two inputs are parallel ($x_i \nparallel x_j$ for $i \neq j$), which is generic for distinct normalized data and is the condition identified by [2]. The matrix H^∞ is exactly the Neural Tangent Kernel Gram matrix induced by the first-layer ReLU features at infinite width.

We record two standard external facts used repeatedly below.

Fact 1.4 (Hoeffding’s inequality, [3, Thm. 2.2.6]). Let $Z_1, \dots, Z_m$ be independent random variables with $Z_r \in [\alpha_r, \beta_r]$ almost surely, and let $S = \sum_{r=1}^m Z_r$. Then for every $t > 0$,

$$\mathbb{P}[|S - \mathbb{E}S| \geq t] \leq 2 \exp\left(-\frac{2t^2}{\sum_{r=1}^m (\beta_r - \alpha_r)^2}\right).$$

Fact 1.5 (Weyl’s inequality, [1, Cor. III.2.6]). Let $A, B \in \mathbb{R}^{n \times n}$ be symmetric. Then for every $i \in [n]$, $|\lambda_i(A) - \lambda_i(B)| \leq \|A - B\|_2$, where eigenvalues are sorted in decreasing order. In particular $\lambda_{\min}(A) \geq \lambda_{\min}(B) - \|A - B\|_2$.

2 Initialization: the empirical Gram matrix concentrates

Lemma 2.1 (Initial Gram close to population). *Fix $\delta \in (0, 1)$. There is a universal constant $C > 0$ such that if*

$$m \geq \frac{C n^2 \log(n/\delta)}{\lambda_0^2}, \tag{6}$$

then with probability at least $1 - \delta$ over the random initialization $W(0)$,

$$\|H(0) - H^\infty\|_2 \leq \frac{\lambda_0}{4} \quad \text{and consequently} \quad \lambda_{\min}(H(0)) \geq \frac{3}{4} \lambda_0. \tag{7}$$

Proof. The strategy is a four-link chain: control each entry $H_{ij}(0)$ by Hoeffding’s inequality (Fact 1.4), union over the n^2 entries to bound every entrywise deviation simultaneously, pass from the entrywise bound to operator norm via $\|\cdot\|_2 \leq \|\cdot\|_F$, and convert the operator-norm bound into an eigenvalue bound by Weyl’s inequality (Fact 1.5).

Step 1 (entrywise Hoeffding). Fix $i, j \in [n]$ and write $H_{ij}(0) = \frac{1}{m} \sum_{r=1}^m Z_r^{ij}$ with $Z_r^{ij} := x_i^\top x_j \mathbf{1}\{w_r(0)^\top x_i \geq 0\} \mathbf{1}\{w_r(0)^\top x_j \geq 0\}$. The Z_r^{ij} are i.i.d. across r (the $w_r(0)$ are i.i.d.), bounded by $|Z_r^{ij}| \leq |x_i^\top x_j| \leq 1$ (Cauchy-Schwarz with $\|x_i\| = 1$), so $Z_r^{ij} \in [-1, 1]$, and $\mathbb{E}[Z_r^{ij}] = H_{ij}^\infty$ by Definition 1.1. Applying Hoeffding’s inequality (Fact 1.4) to $S = \sum_{r=1}^m Z_r^{ij}$ with $\beta_r - \alpha_r = 2$ and threshold $t = m\tau$ for $\tau := \lambda_0/(4n)$,

$$\mathbb{P}[|H_{ij}(0) - H_{ij}^\infty| \geq \tau] = \mathbb{P}[|S - \mathbb{E}S| \geq m\tau] \leq 2 \exp\left(-\frac{2(m\tau)^2}{4m}\right) = 2 \exp\left(-\frac{m\tau^2}{2}\right), \tag{8}$$

where the equality rescales by $1/m$, the inequality is Fact 1.4 with $\sum_r (\beta_r - \alpha_r)^2 = 4m$, and the last is arithmetic.

Step 2 (union over entries). Union-bounding Eq. (8) over all n^2 ordered pairs (i, j) ,

$$\mathbb{P}\left[\max_{i,j \in [n]} |H_{ij}(0) - H_{ij}^\infty| \geq \tau\right] \leq n^2 \cdot 2 \exp\left(-\frac{m\tau^2}{2}\right) \leq \delta, \quad (9)$$

where the first step is the union bound over the n^2 events of Eq. (8) and the second holds whenever $m \geq \frac{2}{\tau^2} \log(2n^2/\delta) = \frac{32n^2}{\lambda_0^2} \log(2n^2/\delta)$; since $\log(2n^2/\delta) \leq 4 \log(n/\delta)$ for $n \geq 1$, this is implied by the width condition Eq. (6) with $C \geq 128$.

Step 3 (entrywise $\geq$ Frobenius $\geq$ operator). On the complementary event of Eq. (9), which has probability at least $1 - \delta$, every entry satisfies $|H_{ij}(0) - H_{ij}^\infty| \leq \tau = \lambda_0/(4n)$, so

$$\|H(0) - H^\infty\|_2 \leq \|H(0) - H^\infty\|_F = \left(\sum_{i,j} |H_{ij}(0) - H_{ij}^\infty|^2\right)^{1/2} \leq \sqrt{n^2 \tau^2} = n\tau = \frac{\lambda_0}{4}, \quad (10)$$

where the first step is the standard bound $\|M\|_2 \leq \|M\|_F$ for any matrix M (the largest singular value is at most the root-sum-of-squares of all singular values), the second is the definition of the Frobenius norm, the third inserts the entrywise bound from Eq. (9) into each of the n^2 summands, and the rest is arithmetic.

Step 4 (Weyl). Applying Weyl's inequality (Fact 1.5) to the symmetric matrices $H(0)$ and H^∞ , together with $\lambda_{\min}(H^\infty) = \lambda_0$ (Assumption 1.2),

$$\lambda_{\min}(H(0)) \geq \lambda_{\min}(H^\infty) - \|H(0) - H^\infty\|_2 \geq \lambda_0 - \frac{\lambda_0}{4} = \frac{3}{4} \lambda_0, \quad (11)$$

where the first step is Weyl's inequality, the second substitutes Assumption 1.2 and Eq. (10), and the last is arithmetic. This proves Eq. (7), completing the proof. $\square$

3 Stability: the Gram matrix survives small perturbations

Lemma 3.1 (Gram stability under weight perturbation). *Fix $\delta \in (0, 1)$ and a radius*

$$R = \frac{c\delta\lambda_0}{n^2}, \quad (12)$$

for a sufficiently small universal constant $c > 0$. Let $W = (w_1, \dots, w_m)$ be any weights (possibly depending on the data) with $\|w_r - w_r(0)\| \leq R$ for every $r \in [m]$. Then with probability at least $1 - \delta$ over the random initialization $W(0)$,

$$\|H(W) - H(0)\|_2 \leq \frac{\lambda_0}{4}. \quad (13)$$

Consequently, on the intersection with the event of Lemma 2.1, $\lambda_{\min}(H(W)) \geq \frac{1}{2}\lambda_0$.

Proof. The strategy is: identify the event that lets a neuron change the Gram matrix (a sign flip of its activation), bound its probability by Gaussian anti-concentration, take expectation to control $\mathbb{E}\|H(W) - H(0)\|_2$, and convert to high probability by Markov.

Step 1 (a neuron contributes only if it flips). With σ the ReLU, $\mathbf{1}\{w_r^\top x_i \geq 0\}$ is the activation indicator of neuron r on x_i . Subtracting Eq. (4) at W and at $W(0)$,

$$|H_{ij}(W) - H_{ij}(0)| \leq \frac{1}{m} \sum_{r=1}^m \left| x_i^\top x_j \right| \cdot |S_{r,ij}(W) - S_{r,ij}(0)| \quad (14)$$

$$\leq \frac{1}{m} \sum_{r=1}^m \mathbf{1}\{E_{r,i} \cup E_{r,j}\}, \quad (15)$$

where $S_{r,ij}(\cdot) := \mathbf{1}\{w_r^\top x_i \geq 0\} \mathbf{1}\{w_r^\top x_j \geq 0\}$, the first step is the triangle inequality applied termwise, and the second uses $|x_i^\top x_j| \leq 1$ together with the observation that $S_{r,ij}(W) \neq S_{r,ij}(0)$ forces neuron r to flip its sign on x_i or on x_j ; here

$$E_{r,i} := \left\{ \left| w_r(0)^\top x_i \right| \leq R \right\}. \quad (16)$$

The inclusion holds because, by Cauchy–Schwarz and $\|x_i\| = 1$, $|(w_r - w_r(0))^\top x_i| \leq \|w_r - w_r(0)\| \leq R$, so a sign change of $w_r^\top x_i$ relative to $w_r(0)^\top x_i$ requires $|w_r(0)^\top x_i| \leq R$.

Step 2 (Gaussian anti-concentration). Since $\|x_i\| = 1$, the projection $w_r(0)^\top x_i \sim \mathcal{N}(0, 1)$. The standard normal density is bounded by $1/\sqrt{2\pi} \leq 1/2$, so for the small-ball event Eq. (16),

$$\mathbb{P}[E_{r,i}] = \mathbb{P}[|\mathcal{N}(0, 1)| \leq R] = \int_{-R}^R \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt \leq \frac{2R}{\sqrt{2\pi}} \leq R, \quad (17)$$

where the first equality is the definition of $E_{r,i}$, the second writes the Gaussian probability as an integral, the third bounds the integrand by its maximum $1/\sqrt{2\pi}$ over the interval of length $2R$, and the last uses $2/\sqrt{2\pi} \leq 1$.

Step 3 (a W -independent dominating quantity). The right-hand side of Eq. (14) depends on $W(0)$ alone, not on W , and dominates $|H_{ij}(W) - H_{ij}(0)|$ for *every* W in the ball simultaneously. Define the single random variable

$$D := \sum_{i=1}^n \sum_{j=1}^n \frac{1}{m} \sum_{r=1}^m \mathbf{1}\{E_{r,i} \cup E_{r,j}\}, \quad (18)$$

which is a function of $W(0)$ only. By Eq. (14) and $\|\cdot\|_2 \leq \|\cdot\|_F \leq \sum_{ij} |\cdot|$,

$$\sup_{W: \|w_r - w_r(0)\| \leq R \forall r} \|H(W) - H(0)\|_2 \leq D. \quad (19)$$

Taking expectations and using the union bound $\mathbb{P}[E_{r,i} \cup E_{r,j}] \leq \mathbb{P}[E_{r,i}] + \mathbb{P}[E_{r,j}] \leq 2R$ together with Eq. (17) for each of the m neurons,

$$\mathbb{E}[D] = \sum_{i=1}^n \sum_{j=1}^n \frac{1}{m} \sum_{r=1}^m \mathbb{P}[E_{r,i} \cup E_{r,j}] \leq \sum_{i=1}^n \sum_{j=1}^n 2R = 2n^2 R, \quad (20)$$

where the first step is linearity of expectation, the second applies Eq. (17) to each neuron, and the last counts the n^2 entries.

Step 4 (Markov to high probability, uniformly over W). By Markov's inequality applied to the nonnegative W -independent random variable D ,

$$\mathbb{P}\left[D \geq \frac{\lambda_0}{4}\right] \leq \frac{4}{\lambda_0} \mathbb{E}[D] \leq \frac{8n^2 R}{\lambda_0} = 8c\delta \leq \delta, \quad (21)$$

where the first step is Markov, the second substitutes Eq. (20), the third substitutes the radius Eq. (12), and the last holds for $c \leq 1/8$. On the complementary event $\{D \leq \lambda_0/4\}$, which has probability at least $1 - \delta$ and depends on $W(0)$ alone, $\|H(W) - H(0)\|_2 \leq D \leq \lambda_0/4$ holds for *every* admissible W at once; in particular it applies to the data-dependent iterates $W(k)$ downstream. This establishes Eq. (13) as the uniform event claimed.

Step 5 (eigenvalue floor). On the intersection with the event of Lemma 2.1 (where $\lambda_{\min}(H(0)) \geq \frac{3}{4}\lambda_0$), Weyl's inequality (Fact 1.5) gives

$$\lambda_{\min}(H(W)) \geq \lambda_{\min}(H(0)) - \|H(W) - H(0)\|_2 \geq \frac{3}{4}\lambda_0 - \frac{1}{4}\lambda_0 = \frac{1}{2}\lambda_0, \quad (22)$$

where the first step is Weyl applied to the symmetric matrices $H(W)$ and $H(0)$ and the second substitutes the two events. This gives the assertion. $\square$

4 Contraction: one gradient step shrinks the residual

Lemma 4.1 (One-step contraction and weight movement). *Consider a single gradient-descent step Eq. (3) from $W(k)$ to $W(k+1)$. Suppose the step size satisfies $\eta \leq \lambda_0/(2n^2)$. Then the per-neuron movement is bounded by*

$$\|w_r(k+1) - w_r(k)\| \leq \frac{\eta\sqrt{n}}{\sqrt{m}} \|y - u(k)\|, \quad \forall r \in [m]. \quad (23)$$

If, in addition, the Gram lower bound $\lambda_{\min}(H(k)) \geq \frac{1}{2}\lambda_0$ holds and every neuron keeps its activation pattern on all inputs over this step, then

$$\|y - u(k+1)\|^2 \leq \left(1 - \frac{\eta\lambda_0}{2}\right) \|y - u(k)\|^2. \quad (24)$$

Proof. The strategy is to differentiate the loss to bound the per-neuron gradient (giving Eq. (23)), then expand the squared residual along the step and identify the Gram matrix in the cross term (giving Eq. (24)).

Step 1 (per-neuron gradient). Differentiating Eq. (2) through Eq. (1),

$$\frac{\partial L(W)}{\partial w_r} = \sum_{i=1}^n (u_i - y_i) \frac{\partial u_i}{\partial w_r} = \frac{1}{\sqrt{m}} \sum_{i=1}^n (u_i - y_i) a_r \mathbf{1}\{w_r^\top x_i \geq 0\} x_i, \quad (25)$$

where the first step is the chain rule on Eq. (2) and the second substitutes $\partial u_i / \partial w_r = \frac{1}{\sqrt{m}} a_r \sigma'(w_r^\top x_i) x_i$ with $\sigma'(z) = \mathbf{1}\{z \geq 0\}$. Taking norms,

$$\left\| \frac{\partial L(W)}{\partial w_r} \right\| \leq \frac{1}{\sqrt{m}} \sum_{i=1}^n |u_i - y_i| |a_r| \mathbf{1}\{w_r^\top x_i \geq 0\} \|x_i\| \quad (26)$$

$$\leq \frac{1}{\sqrt{m}} \sum_{i=1}^n |u_i - y_i| \leq \frac{\sqrt{n}}{\sqrt{m}} \|y - u\|, \quad (27)$$

where the first step is the triangle inequality, the second uses $|a_r| = 1$, $\mathbf{1}\{\cdot\} \leq 1$ and $\|x_i\| = 1$, and the last is Cauchy-Schwarz $\sum_i |u_i - y_i| \leq \sqrt{n} \|y - u\|$. Multiplying by the step size η in Eq. (3) gives Eq. (23).

Step 2 (residual recursion identifies the Gram matrix). Assume each neuron keeps its activation pattern $\mathbf{1}\{w_r^\top x_i \geq 0\}$ unchanged over the step (so f is locally linear in W along the step). Then for each i ,

$$u_i(k+1) - u_i(k) = \left\langle \frac{\partial u_i}{\partial W}(k), W(k+1) - W(k) \right\rangle = -\eta \left\langle \frac{\partial u_i}{\partial W}(k), \nabla_W L(W(k)) \right\rangle \quad (28)$$

$$= -\eta \sum_{j=1}^n (u_j(k) - y_j) H_{ij}(k), \quad (29)$$

where the first step is the (locally exact) linearity of u_i in W under a fixed activation pattern, the second substitutes the update Eq. (3), and the third uses the Jacobian inner product

$$\left\langle \frac{\partial u_i}{\partial W}, \frac{\partial u_j}{\partial W} \right\rangle = \frac{1}{m} \sum_{r=1}^m a_r^2 x_i^\top x_j \mathbf{1}\{w_r^\top x_i \geq 0\} \mathbf{1}\{w_r^\top x_j \geq 0\} = H_{ij}(k), \quad (30)$$

which follows from Eq. (25) and Eq. (4) with $a_r^2 = 1$. In vector form $u(k+1) - u(k) = -\eta H(k) (u(k) - y)$, hence

$$y - u(k+1) = (I - \eta H(k)) (y - u(k)). \quad (31)$$

Step 3 (contraction in squared norm). Taking squared norms of Eq. (31),

$$\|y - u(k+1)\|^2 = (y - u(k))^\top (I - \eta H(k))^2 (y - u(k)) \quad (32)$$

$$\leq \|I - \eta H(k)\|_2^2 \cdot \|y - u(k)\|^2 \quad (33)$$

$$\leq (1 - \eta \lambda_{\min}(H(k)))^2 \|y - u(k)\|^2 \quad (34)$$

$$\leq \left(1 - \frac{\eta \lambda_0}{2}\right)^2 \|y - u(k)\|^2 \leq \left(1 - \frac{\eta \lambda_0}{2}\right) \|y - u(k)\|^2, \quad (35)$$

where the first step expands the quadratic form, the second is the Rayleigh bound $v^\top M^2 v \leq \|M\|_2^2 \|v\|^2$, the third uses that $I - \eta H(k)$ is symmetric with eigenvalues $1 - \eta \lambda_\ell(H(k)) \in [1 - \eta \lambda_{\max}(H(k)), 1 - \eta \lambda_{\min}(H(k))]$ all lying in $[0, 1 - \eta \lambda_{\min}(H(k))]$ — nonnegativity of the lower endpoint uses $\eta \lambda_{\max}(H(k)) \leq \eta n \leq \frac{\lambda_0}{2n} \leq 1$ since $\lambda_{\max}(H(k)) \leq \|H(k)\|_2 \leq n$ and $\eta \leq \lambda_0/(2n^2)$ — so $\|I - \eta H(k)\|_2 \leq 1 - \eta \lambda_{\min}(H(k))$; the fourth substitutes $\lambda_{\min}(H(k)) \geq \frac{1}{2} \lambda_0$; and the last uses $(1 - x)^2 \leq 1 - x$ for $x \in [0, 1]$ with $x = \frac{\eta \lambda_0}{2} \in [0, 1]$. This is Eq. (24), completing the proof. $\square$

5 Fixed-point induction: the iterates stay in the basin

The three preceding lemmas hold each at a fixed configuration. The work of this section is to weld them into a statement that holds *simultaneously for all k* , by a fixed-point / contradiction argument: linear convergence keeps the weights in a ball of radius R , and staying in that ball keeps the Gram matrix conditioned, which in turn drives the linear convergence.

Lemma 5.1 (Trajectory stays in the basin). *Fix $\delta \in (0, 1)$ and let $R = c\delta\lambda_0/n^2$ be the radius of Lemma 3.1. Suppose the following conditions hold:*

- the step size $\eta = \kappa \lambda_0/n^2$ for a universal constant $\kappa \leq 1/2$;
- the width satisfies

$$m \geq C \cdot \frac{n^6}{\lambda_0^4 \delta^3} \quad \text{for a sufficiently large universal constant } C > 0, \quad (36)$$

- the events of Lemma 2.1 and Lemma 3.1 both hold, the initial-residual event of Lemma 5.2 (used in part (a) below) holds, together with the flip-count event $\mathcal{E}_4$ of Eq. (46) below (a function of $W(0)$ alone); their intersection has probability at least $1 - 4\delta$ over the random initialization.

Then for every $k \geq 0$,

$$\|y - u(k)\|^2 \leq \left(1 - \frac{\eta\lambda_0}{2}\right)^k \|y - u(0)\|^2, \quad (37)$$

$$\|w_r(k) - w_r(0)\| \leq R \quad \forall r \in [m], \quad (38)$$

$$\lambda_{\min}(H(k)) \geq \frac{1}{2}\lambda_0. \quad (39)$$

Proof. We argue by strong induction on k , proving the conjunction $P(k)$: “Eq. (37), Eq. (38) and Eq. (39) hold at index k ”. Throughout we work on the intersection of the four high-probability events (Lemma 2.1, Lemma 3.1, the initial-residual event of Lemma 5.2, and the flip-count event $\mathcal{E}_4$ of Eq. (46)), on which all statements below are deterministic.

Base case $k = 0$. Eq. (37) is an equality at $k = 0$; Eq. (38) is trivial since $w_r(0) - w_r(0) = 0 \leq R$; Eq. (39) is exactly Lemma 2.1, which gives $\lambda_{\min}(H(0)) \geq \frac{3}{4}\lambda_0 \geq \frac{1}{2}\lambda_0$.

Inductive step. Assume $P(0), \dots, P(k)$. We first upgrade the stay-in-ball bound to index $k + 1$, then use it to maintain the Gram floor and the contraction.

(a) *Weight movement at $k + 1$.* Summing the one-step movement Eq. (23) of Lemma 4.1 (whose step-size hypothesis $\eta \leq \lambda_0/(2n^2)$ holds since $\kappa \leq 1/2$) along $s = 0, \dots, k$, and inserting the inductive convergence Eq. (37),

$$\|w_r(k+1) - w_r(0)\| \leq \sum_{s=0}^k \|w_r(s+1) - w_r(s)\| \leq \sum_{s=0}^k \frac{\eta\sqrt{n}}{\sqrt{m}} \|y - u(s)\| \quad (40)$$

$$\leq \frac{\eta\sqrt{n}}{\sqrt{m}} \|y - u(0)\| \sum_{s=0}^{\infty} \left(1 - \frac{\eta\lambda_0}{2}\right)^{s/2} \leq \frac{\eta\sqrt{n}}{\sqrt{m}} \|y - u(0)\| \cdot \frac{4}{\eta\lambda_0} \quad (41)$$

$$= \frac{4\sqrt{n}}{\sqrt{m}\lambda_0} \|y - u(0)\|, \quad (42)$$

where the first step is the triangle inequality over the telescoping path, the second applies Eq. (23) at each s , the third inserts Eq. (37) and extends the sum to infinity, the fourth uses the geometric-series identity and bound $\sum_{s \geq 0} (1 - \frac{\eta\lambda_0}{2})^{s/2} = \frac{1}{1 - \sqrt{1 - \eta\lambda_0/2}} \leq \frac{4}{\eta\lambda_0}$ (the equality sums the geometric series with ratio $\sqrt{1 - \eta\lambda_0/2} < 1$; the inequality uses $1 - \sqrt{1 - x} \geq x/2$ for $x \in [0, 1]$ with $x = \frac{\eta\lambda_0}{2}$, giving $\frac{1}{1 - \sqrt{1 - x}} \leq \frac{2}{x} = \frac{4}{\eta\lambda_0}$), and the last cancels η . With the initial-residual bound $\|y - u(0)\| \leq \sqrt{2n/\delta}$ (Lemma 5.2; when the theorem invokes this lemma at level $\delta/4$ the bound reads $\sqrt{8n/\delta}$, which only enlarges the constant C below and is absorbed into it), the right-hand side is at most $4\sqrt{2}n/(\sqrt{m}\delta\lambda_0)$, which is $\leq R = c\delta\lambda_0/n^2$ precisely when $m \geq 32n^6/(c^2\delta^3\lambda_0^4)$; this is Eq. (36) with $C \geq 32/c^2$. Hence Eq. (38) holds at $k + 1$.

(b) *Gram floor at $k + 1$.* Since Eq. (38) holds at $k + 1$, every neuron satisfies $\|w_r(k+1) - w_r(0)\| \leq R$, so Lemma 3.1 applies at $W(k+1)$ and yields $\|H(k+1) - H(0)\|_2 \leq \frac{\lambda_0}{4}$; with $\lambda_{\min}(H(0)) \geq \frac{3}{4}\lambda_0$ and Weyl’s inequality (Fact 1.5) this gives

$$\lambda_{\min}(H(k+1)) \geq \lambda_{\min}(H(0)) - \|H(k+1) - H(0)\|_2 \geq \frac{3}{4}\lambda_0 - \frac{1}{4}\lambda_0 = \frac{1}{2}\lambda_0, \quad (43)$$

which is Eq. (39) at $k + 1$.

(c) *Contraction from k to $k + 1$.* The Gram floor $\lambda_{\min}(H(k)) \geq \frac{1}{2}\lambda_0$ (Eq. (39) at k) and the step-size cap meet the hypotheses of Lemma 4.1. It remains to discharge its activation-pattern hypothesis: the recursion Eq. (31) is the fixed-pattern linearization, and neuron–input pairs that flip over the step contribute a remainder $\varepsilon(k)$. Let $\mathcal{S}_k \subseteq [m] \times [n]$ be the set of flipping pairs — those (r, i) for

which neuron r changes its activation on input x_i over the step — and $R_{\text{step}} := \frac{\eta\sqrt{n}}{\sqrt{m}} \|y - u(k)\|$ the per-step movement Eq. (23). A pair (r, i) flips over the step only if

$$\left| w_r(0)^\top x_i \right| \leq \|w_r(k) - w_r(0)\| + \|w_r(k+1) - w_r(k)\| \leq R + R_{\text{step}} \leq 2R, \quad (44)$$

where the first step is Cauchy–Schwarz, the second uses the inductive stay-in-ball bound Eq. (38) at k and the movement Eq. (23), and the last uses $R_{\text{step}} \leq R$ for $m \geq Cn^6/(\lambda_0^4\delta^3)$. The flip event is thus anchored at the *Gaussian* point $w_r(0) \sim \mathcal{N}(0, 1)$, so anti-concentration Eq. (17) (with $R \mapsto 2R$) gives

$$\mathbb{E} |\mathcal{S}_k| \leq \sum_{r=1}^m \sum_{i=1}^n \mathbb{P} \left[\left| w_r(0)^\top x_i \right| \leq 2R \right] \leq \sum_{r=1}^m \sum_{i=1}^n 2R = 2nmR, \quad (45)$$

where the first step writes $|\mathcal{S}_k| = \sum_{r,i} \mathbf{1}\{(r, i) \text{ flips}\}$ and bounds each flip probability by the anchor event Eq. (44), the second applies Eq. (17) at radius $2R$, and the last counts the nm terms. The count $|\mathcal{S}_k|$ is itself random, so to obtain a per-trajectory bound we control it through a single event that does not depend on k . Every flipping pair satisfies the anchor Eq. (44), hence $\mathcal{S}_k \subseteq \mathcal{N}$ for all k , where $\mathcal{N} := \{(r, i) : |w_r(0)^\top x_i| \leq 2R\}$ depends on $W(0)$ alone. By the same count as Eq. (45), $\mathbb{E} |\mathcal{N}| \leq 2nmR$, so Markov at level $\delta/4$ gives the event

$$\mathcal{E}_4 := \left\{ |\mathcal{N}| \leq \frac{4}{\delta} \mathbb{E} |\mathcal{N}| \leq \frac{8nmR}{\delta} \right\}, \quad \mathbb{P}[\mathcal{E}_4^c] \leq \frac{\delta}{4}, \quad (46)$$

on which $|\mathcal{S}_k| \leq 8nmR/\delta$ deterministically and simultaneously for every k . Each flipping pair (r, i) perturbs the i -th coordinate of u by at most $\frac{1}{\sqrt{m}} R_{\text{step}}$, so summing the contributions of the at most $|\mathcal{S}_k|$ flipping pairs (triangle inequality, no cancellation) across the n coordinates,

$$\|\varepsilon(k)\| \leq \frac{\eta n}{m} \cdot |\mathcal{S}_k| \cdot \|y - u(k)\| \leq \frac{\eta n}{m} \cdot \frac{8nmR}{\delta} \cdot \|y - u(k)\| \leq \frac{\eta \lambda_0}{8} \|y - u(k)\|, \quad (47)$$

where the first step aggregates the $|\mathcal{S}_k|$ per-pair perturbations $\frac{1}{\sqrt{m}} R_{\text{step}} = \frac{\eta\sqrt{n}}{m} \|y - u(k)\|$ by the triangle inequality within each coordinate and Cauchy–Schwarz over the n coordinates, the second inserts the event $\mathcal{E}_4$ from Eq. (46), and the third substitutes $R = c\delta\lambda_0/n^2$ (giving prefactor $8c\eta\lambda_0 \leq \eta\lambda_0/8$ for $c \leq 1/64$, with n, m and δ all cancelling). The aggregation is thus driven by the constant c alone, not the width.

The flipping pairs make the exact recursion Eq. (31) hold only up to the remainder $\varepsilon(k)$, i.e. $y - u(k+1) = (I - \eta H(k))(y - u(k)) + \varepsilon(k)$. Combining the contraction Eq. (32) for the linear part with Eq. (47) for the remainder via the triangle inequality,

$$\|y - u(k+1)\| \leq \|(I - \eta H(k))(y - u(k))\| + \|\varepsilon(k)\| \leq \left(1 - \frac{\eta\lambda_0}{2}\right) \|y - u(k)\| + \frac{\eta\lambda_0}{8} \|y - u(k)\| \quad (48)$$

$$= \left(1 - \frac{3\eta\lambda_0}{8}\right) \|y - u(k)\|, \quad (49)$$

where the first step is the triangle inequality on the perturbed recursion, the second bounds the linear part by $\|I - \eta H(k)\|_2 \leq 1 - \eta\lambda_0/2$ (the operator bound from Eq. (32), using $\lambda_{\min}(H(k)) \geq \frac{1}{2}\lambda_0$) and the remainder by Eq. (47). Squaring and using $(1 - \frac{3\eta\lambda_0}{8})^2 \leq 1 - \frac{\eta\lambda_0}{2}$ — valid since $\frac{9}{64}(\eta\lambda_0)^2 \leq \frac{1}{4}\eta\lambda_0$ for $\eta\lambda_0 \leq \frac{16}{9}$, which holds as $\eta\lambda_0 \leq \kappa\lambda_0^2/n^2 \leq \frac{1}{2}$ — then inserting the inductive hypothesis Eq. (37) at k ,

$$\|y - u(k+1)\|^2 \leq \left(1 - \frac{\eta\lambda_0}{2}\right) \|y - u(k)\|^2 \leq \left(1 - \frac{\eta\lambda_0}{2}\right)^{k+1} \|y - u(0)\|^2. \quad (50)$$

This is Eq. (37) at $k + 1$.

Parts (a)–(c) establish $P(k + 1)$. By induction, $P(k)$ holds for all $k \geq 0$, which is the assertion. $\square$

Lemma 5.2 (Initial residual is bounded). *Fix $\delta \in (0, 1)$. With probability at least $1 - \delta$ over the random initialization, $\|y - u(0)\| \leq \sqrt{2n/\delta}$.*

Proof. We bound the second moment and apply Markov. Conditioned on $W(0)$, the output weights a_r are i.i.d. $\text{Unif}\{\pm 1\}$ and independent across r , so $\mathbb{E}[a_r] = 0$ and each $u_i(0) = \frac{1}{\sqrt{m}} \sum_r a_r \sigma(w_r(0)^\top x_i)$ is a mean-zero sum, giving

$$\mathbb{E}[\|y - u(0)\|^2] = \sum_{i=1}^n (y_i^2 + \mathbb{E}[u_i(0)^2]) = \sum_{i=1}^n \left(y_i^2 + \frac{1}{m} \sum_{r=1}^m \mathbb{E}[\sigma(w_r(0)^\top x_i)^2] \right) \quad (51)$$

$$\leq \sum_{i=1}^n \left(1 + \frac{1}{m} \sum_{r=1}^m \mathbb{E}[(w_r(0)^\top x_i)^2] \right) = \sum_{i=1}^n (1 + 1) = 2n, \quad (52)$$

where the first step expands $\|y - u(0)\|^2$ and uses $\mathbb{E}[u_i(0)] = 0$ to kill the cross term, the second uses independence of the a_r ($\mathbb{E}[a_r a_{r'}] = \mathbf{1}\{r = r'\}$) and $a_r^2 = 1$, the third uses $|y_i| \leq 1$ and $\sigma(z)^2 \leq z^2$, and the fourth uses $w_r(0)^\top x_i \sim \mathcal{N}(0, 1)$ so $\mathbb{E}[(w_r(0)^\top x_i)^2] = 1$. Applying Markov's inequality to the nonnegative random variable $\|y - u(0)\|^2$ at threshold $2n/\delta$,

$$\mathbb{P}\left[\|y - u(0)\|^2 \geq \frac{2n}{\delta}\right] \leq \frac{\delta}{2n} \mathbb{E}[\|y - u(0)\|^2] \leq \frac{\delta}{2n} \cdot 2n = \delta, \quad (53)$$

where the first step is Markov and the second substitutes Eq. (51). Taking complements and square roots, $\|y - u(0)\| \leq \sqrt{2n/\delta}$ with probability at least $1 - \delta$. This gives the assertion. $\square$

6 Main theorem: linear convergence to zero training loss

Remark 6.1 (Informal statement). Under Assumption 1.2, suppose the width is $m \gtrsim \text{poly}(n, 1/\lambda_0, 1/\delta)$ and the step size is $\eta = O(\lambda_0/n^2)$. Then with high probability over the random initialization, gradient descent on the squared loss Eq. (2) converges linearly to zero training loss: $\|y - u(k)\|^2 \leq (1 - \frac{\eta\lambda_0}{2})^k \|y - u(0)\|^2 \rightarrow 0$. The formal version with constants pinned is Theorem 6.2.

Theorem 6.2 (Linear convergence of gradient descent). *Let Assumption 1.2 hold and fix $\delta \in (0, 1)$. Suppose*

$$m \geq C \frac{n^6}{\lambda_0^4 \delta^3}, \quad \eta = \kappa \frac{\lambda_0}{n^2} \quad (\kappa \leq \tfrac{1}{2}), \quad (54)$$

for a sufficiently large universal constant $C > 0$. Then with probability at least $1 - \delta$ over the random initialization $\{(w_r(0), a_r)\}_{r=1}^m$, gradient descent Eq. (3) satisfies, for every iteration $k \geq 0$,

$$\|y - u(k)\|^2 \leq \left(1 - \frac{\eta\lambda_0}{2}\right)^k \|y - u(0)\|^2. \quad (55)$$

In particular $\|y - u(k)\|^2 \rightarrow 0$ as $k \rightarrow \infty$, so the training loss decays to zero at the linear rate $1 - \frac{\eta\lambda_0}{2} = 1 - \frac{\kappa\lambda_0^2}{2n^2}$, and reaching $\|y - u(k)\|^2 \leq \varepsilon$ takes $k = O\left(\frac{n^2}{\lambda_0^2} \log \frac{n}{\delta\varepsilon}\right)$ steps.

Proof of Theorem 6.2. The proof is an assembly of the four lemmas via a union bound over their failure events.

Union bound (probability budget). Define the four initialization events, each invoked with failure parameter $\delta/4$:

$$\begin{aligned}\mathcal{E}_1 &:= \left\{ \|H(0) - H^\infty\|_2 \leq \frac{\lambda_0}{4} \right\} \quad (\text{Lemma 2.1}), \\ \mathcal{E}_2 &:= \left\{ \|H(W) - H(0)\|_2 \leq \frac{\lambda_0}{4} \text{ whenever } \|w_r - w_r(0)\| \leq R \ \forall r \right\} \quad (\text{Lemma 3.1}), \\ \mathcal{E}_3 &:= \left\{ \|y - u(0)\| \leq \sqrt{8n/\delta} \right\} \quad (\text{Lemma 5.2 at level } \delta/4), \\ \mathcal{E}_4 &:= \left\{ \left| \{(r, i) : |w_r(0)^\top x_i| \leq 2R\} \right| \leq \frac{8nmR}{\delta} \right\} \quad (\text{Eq. (46) at level } \delta/4).\end{aligned}\tag{56}$$

By Lemmas 2.1, 3.1 and 5.2 and the Markov bound Eq. (46), each applied with failure probability $\delta/4$ (the width Eq. (54) dominates each lemma's requirement with δ replaced by $\delta/4$, which only changes the universal constant C), together with a union bound over the four failure events,

$$\mathbb{P}[\mathcal{E}_1 \cap \mathcal{E}_2 \cap \mathcal{E}_3 \cap \mathcal{E}_4] \geq 1 - \mathbb{P}[\mathcal{E}_1^c] - \mathbb{P}[\mathcal{E}_2^c] - \mathbb{P}[\mathcal{E}_3^c] - \mathbb{P}[\mathcal{E}_4^c] \geq 1 - 4 \cdot \frac{\delta}{4} = 1 - \delta, \tag{57}$$

where the first step is the union bound (equivalently $\mathbb{P}[\cap_i \mathcal{E}_i] = 1 - \mathbb{P}[\cup_i \mathcal{E}_i^c]$) and the second inserts the per-lemma failure probabilities.

Deterministic conclusion on $\mathcal{E}_1 \cap \mathcal{E}_2 \cap \mathcal{E}_3 \cap \mathcal{E}_4$. On this event all hypotheses of Lemma 5.1 are met: its three bulleted conditions are exactly the step-size choice, the events $\mathcal{E}_1 \cap \mathcal{E}_2 \cap \mathcal{E}_4$, and (feeding the stay-in-ball width computation) the initial-residual event $\mathcal{E}_3$. Therefore Lemma 5.1 yields Eq. (37) for every $k \geq 0$, which is exactly Eq. (55). Since $0 < 1 - \frac{\eta\lambda_0}{2} < 1$, the right-hand side tends to 0, giving $\|y - u(k)\|^2 \rightarrow 0$. Finally, solving $(1 - \frac{\eta\lambda_0}{2})^k \cdot (8n/\delta) \leq \varepsilon$ and using $-\log(1 - x) \geq x$ gives $k = O(\frac{1}{\eta\lambda_0} \log(\frac{n}{\delta\varepsilon})) = O(\frac{n^2}{\lambda_0^2} \log(\frac{n}{\delta\varepsilon}))$. This completes the proof. $\square$

References

- [1] Rajendra Bhatia. *Matrix Analysis*, volume 169 of *Graduate Texts in Mathematics*. Springer, 1997.
- [2] Simon S. Du, Xiyu Zhai, Barnabas Poczos, and Aarti Singh. Gradient descent provably optimizes over-parameterized neural networks. In *International Conference on Learning Representations (ICLR)*, 2019. arXiv:1810.02054.
- [3] Roman Vershynin. *High-Dimensional Probability: An Introduction with Applications in Data Science*. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, 2018.